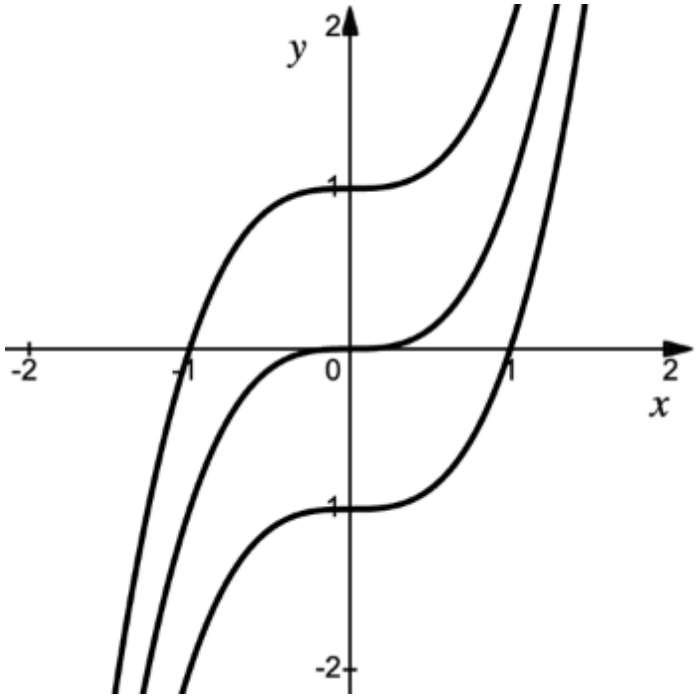
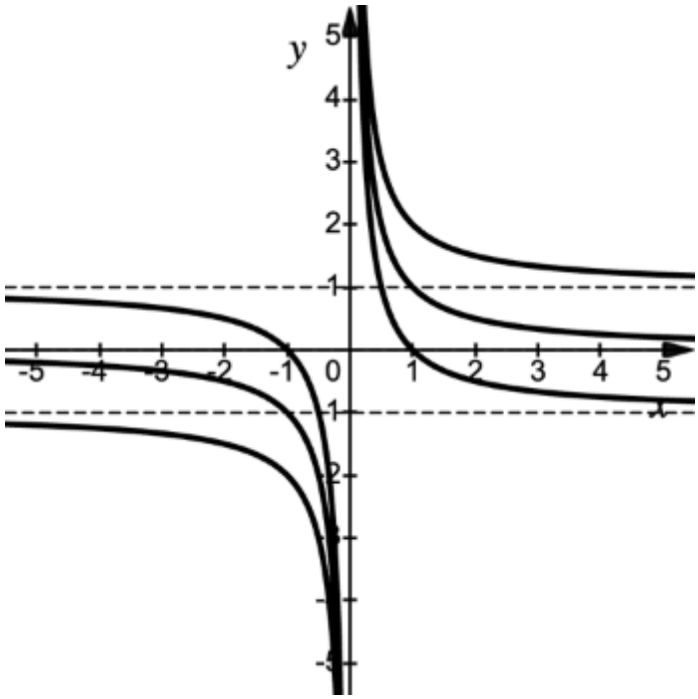
2.40 1st Order DEs: Sketching Solutions

Worked Solutions

1)



2)



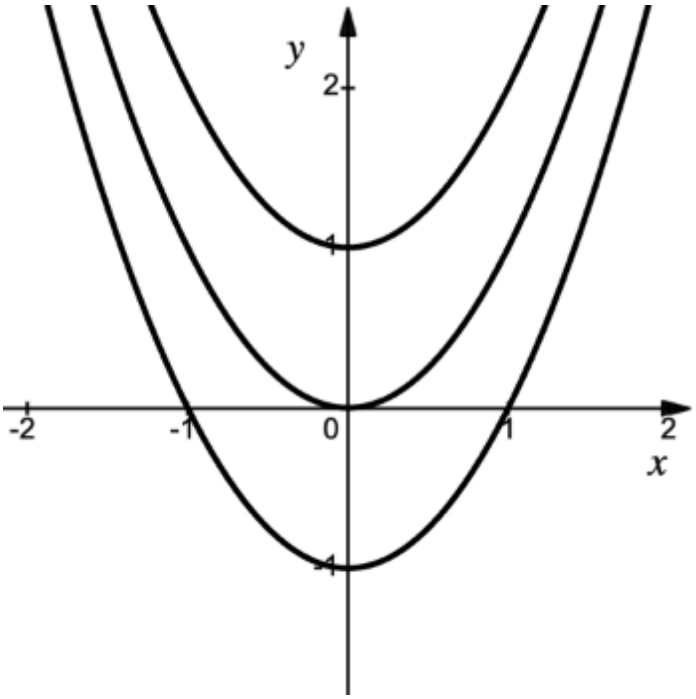
3 a)

The gradient is zero when $\frac{dy}{dx} = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$

3 b)

$$\frac{dy}{dx} = 2x$$
$$\Rightarrow y = x^2 + c$$

3 c)



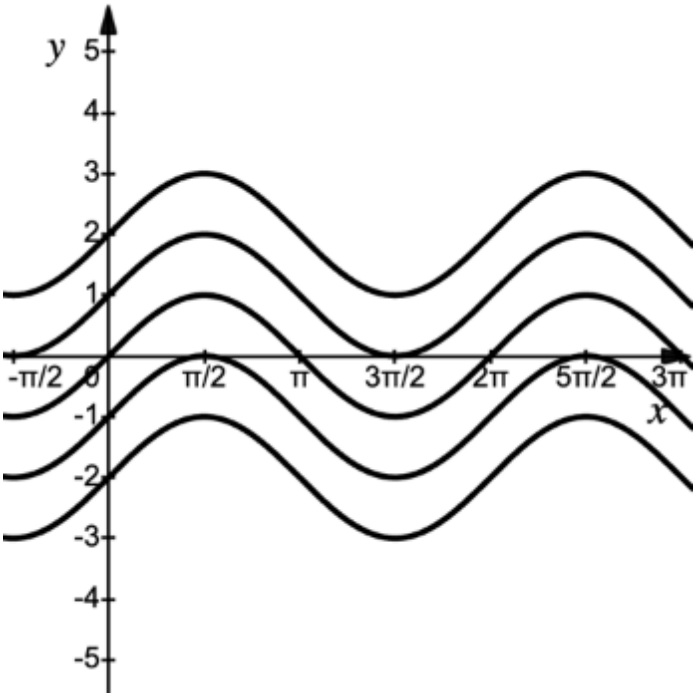
3 d)

The family of solutions are vertical translations of $y = x^2$

4 a)

$$\frac{dy}{dx} = \cos x$$
$$\Rightarrow y = \sin x + c$$

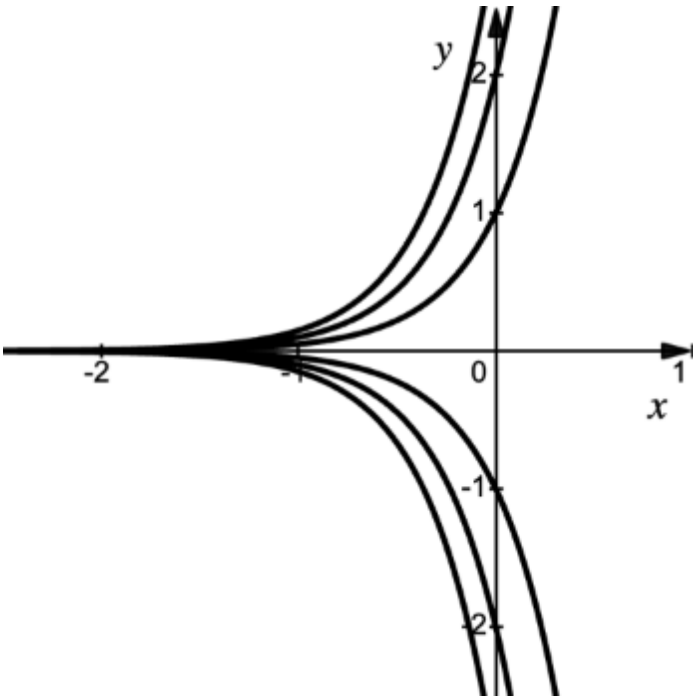
4 b)



5 a)

$$\frac{dy}{dx} = 3y$$
$$\Rightarrow \int \frac{1}{y}\, dy = \int 3\, dx$$
$$\Rightarrow \ln|y| = 3x + c$$
$$\Rightarrow y = \pm e^{3x+c}$$
$$\Rightarrow y = \pm e^{3x} e^{c}$$
$$\Rightarrow y = Ae^{3x}$$

5 b)



5 c)

As $x \to -\infty, y \to 0$ (the x-axis is a horizontal asymptote)

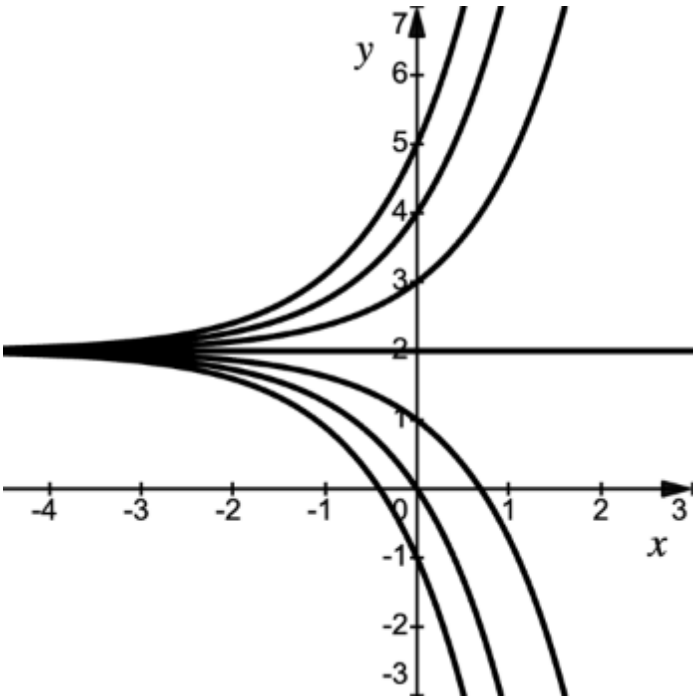
6 a)

When $y = 2$, $\frac{dy}{dx} = 2 - 2 = 0$ and so is a constant solution to the differential equation.

6 b)

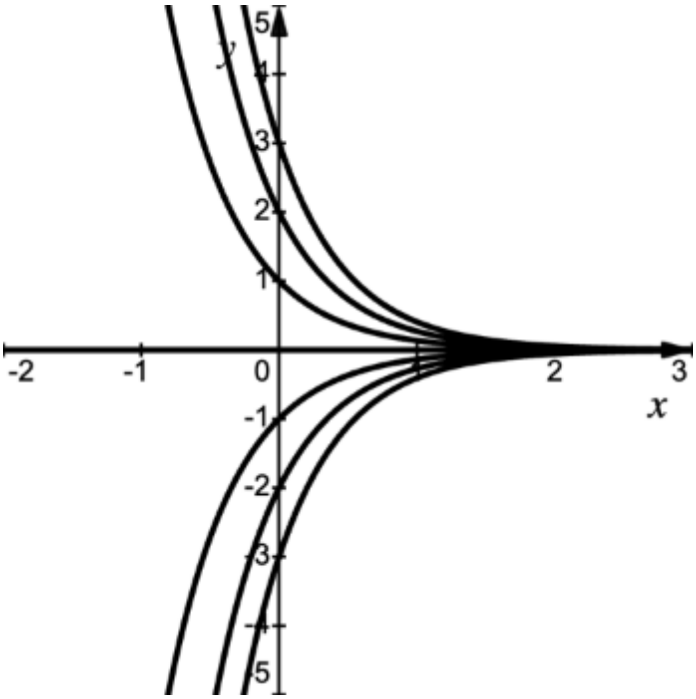
$$\begin{aligned} \frac{dy}{dx} &= y - 2 \\ \Rightarrow \int \frac{1}{y-2}\, dy &= \int dx \\ \Rightarrow \ln|y-2| &= x + c \\ \Rightarrow y - 2 &= \pm e^{x+c} \\ \Rightarrow y &= 2 \pm e^x e^c \\ \Rightarrow y &= 2 + Ae^x \end{aligned}$$

6 c)



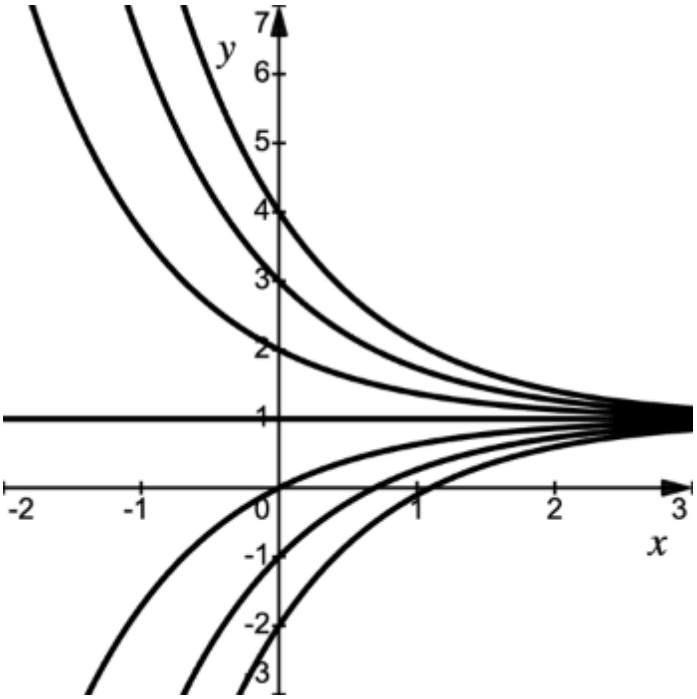
7 a)

$$\frac{dy}{dx} = -2y$$
$$\Rightarrow \int \frac{1}{y}\, dy = \int -2\, dx$$
$$\Rightarrow \ln|y| = -2x + c$$
$$\Rightarrow y = \pm e^{-2x+c}$$
$$\Rightarrow y = \pm e^{-2x}e^{c}$$
$$\Rightarrow y = Ae^{-2x}$$



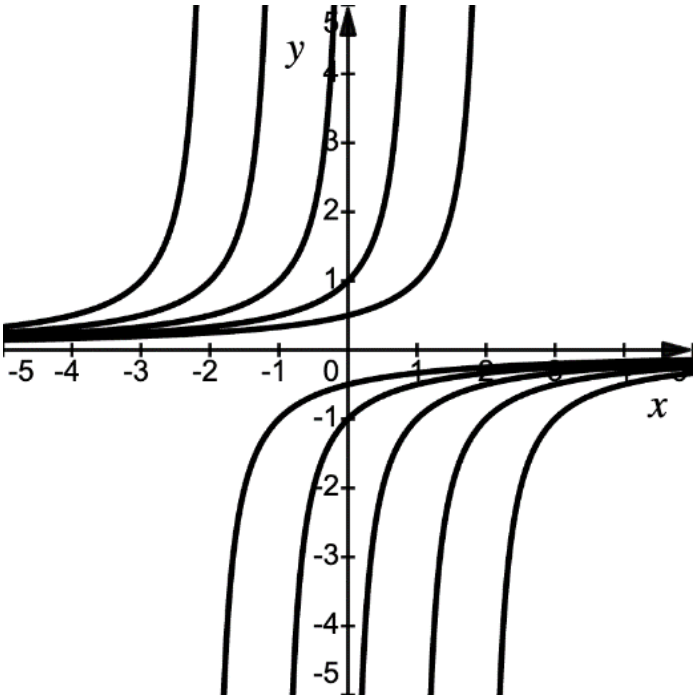
7 b)

$$\frac{dy}{dx} = 1 - y$$
$$\Rightarrow \int \frac{1}{1-y}\, dy = \int dx$$
$$\Rightarrow -\ln|1-y| = x + c$$
$$\Rightarrow \ln|1-y| = k - x$$
$$\Rightarrow 1 - y = \pm e^{k-x}$$
$$\Rightarrow y = 1 \pm e^{k} e^{-x}$$
$$\Rightarrow y = 1 + Ae^{-x}$$



7 c)

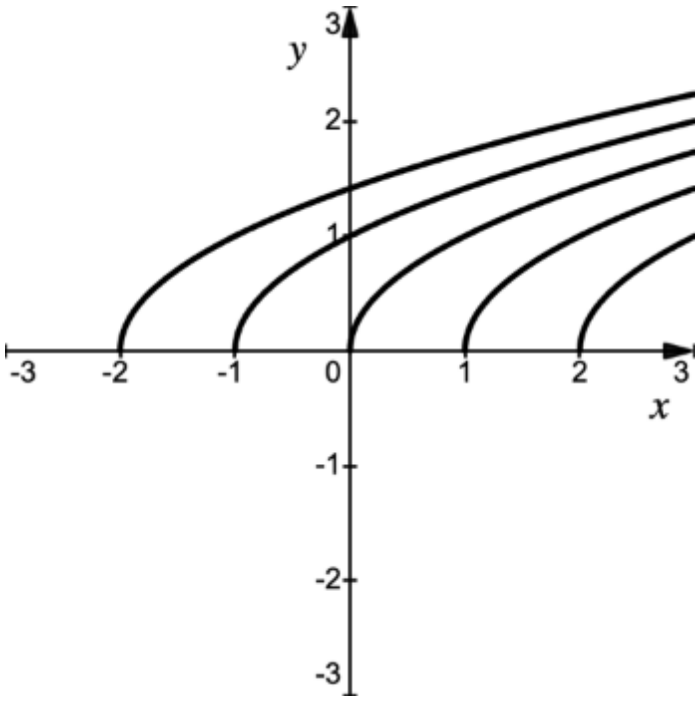
$$\frac{dy}{dx} = y^2$$
$$\Rightarrow \int \frac{1}{y^2}\, dy = \int dx$$
$$\Rightarrow -\frac{1}{y} = x + c$$
$$\Rightarrow \frac{1}{y} = A - x$$
$$\Rightarrow y = \frac{1}{A - x}$$



7 d)

$$\frac{dy}{dx} = \frac{1}{2y}$$
$$\Rightarrow \int 2y\, dy = \int dx$$
$$\Rightarrow y^2 = x + c$$
$$\Rightarrow y = \pm\sqrt{x+c}$$

Given that $y > 0, y = \sqrt{x+c}$



8 a)

When $y = 0$, $\frac{dy}{dx} = 0(1-0) = 0$ and when $y = 1$, $\frac{dy}{dx} = 1(1-1) = 0$

So $y = 0$ and $y = 1$ are constant solutions to the differential equation.

8 b)

$$\frac{dy}{dx} = y(1-y)$$
$$\Rightarrow \int \frac{1}{y(1-y)}\, dy = \int dx$$

Use Partial Fractions:

$$\frac{1}{y(1-y)} \equiv \frac{A}{y} + \frac{B}{1-y}$$
$$\Rightarrow 1 \equiv A(1-y) + By$$

Let $y = 0 : 1 = A$

Let $y = 1 : 1 = B$

$$\begin{aligned}
\therefore \frac{1}{y(1-y)} &\equiv \frac{1}{y} + \frac{1}{1-y} \\
\Rightarrow \int \left(\frac{1}{y} + \frac{1}{1-y}\right) dy &= \int dx \\
\Rightarrow \ln|y| - \ln|1-y| &= x + c \\
\Rightarrow \ln\left|\frac{y}{1-y}\right| &= x + c \\
\Rightarrow \frac{y}{1-y} &= \pm e^{x+c} \\
\Rightarrow \frac{y}{1-y} &= \pm e^x e^c \\
\Rightarrow \frac{y}{1-y} &= Ae^x \\
\Rightarrow y &= Ae^x - Ae^x y \\
\Rightarrow y + Ae^x y &= Ae^x \\
\Rightarrow (1 + Ae^x)y &= Ae^x \\
\Rightarrow y &= \frac{Ae^x}{1 + Ae^x}
\end{aligned}$$

8 c)

In part a), we found $y = 0$ and $y = 1$ are constant solutions of the differential equation.

If $y = 0$:

$$0 = \frac{Ae^x}{1 + Ae^x} \Rightarrow A = 0$$

If $y = 1$:

$$1 = \frac{Ae^x}{1 + Ae^x} \Rightarrow 1 + Ae^x = Ae^x \Rightarrow 1 = 0$$

So $y = 1$ does not appear in the solution found in part b), but $y = 0$ does.

When we separate the variables from $\frac{dy}{dx} = y(1-y)$ to $\int \frac{1}{y(1-y)}\, dy = \int dx$ we are assuming $y \neq 0$ and $y \neq 1$

8 d)

$$\begin{aligned}
y = \frac{Ae^x}{1 + Ae^x} &= \frac{\frac{Ae^x}{Ae^x}}{\frac{1}{Ae^x} + \frac{Ae^x}{Ae^x}} \\
&= \frac{1}{\frac{1}{A}e^{-x} + 1} = \frac{1}{1 + Be^{-x}}
\end{aligned}$$

8 e)

In part a), we found $y = 0$ and $y = 1$ are constant solutions of the differential equation.

If $y = 0$:

$$0 = \frac{1}{1 + Be^{-x}} \Rightarrow 0 = 1$$

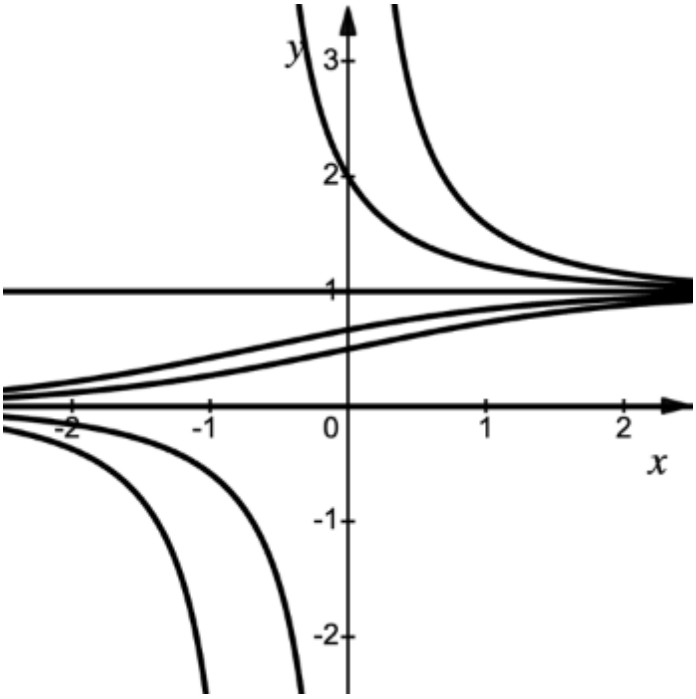
If $y = 1$:

$$1 = \frac{1}{1 + Be^{-x}} \Rightarrow 1 + Be^{-x} = 1 \Rightarrow B = 0$$

So $y = 0$ does not appear in the solution found in part d), but $y = 1$ does.

When we separate the variables from $\frac{dy}{dx} = y(1-y)$ to $\int \frac{1}{y(1-y)}\,dy = \int dx$ we are assuming $y \neq 0$ and $y \neq 1$

8 f)



9 a)

Trial:

$$y = k \Rightarrow \frac{dy}{dx} = 0$$
$$\therefore 0 + k = 3 \Rightarrow k = 3$$

So $y = 3$ is a constant solution of the differential equation.

9 b)

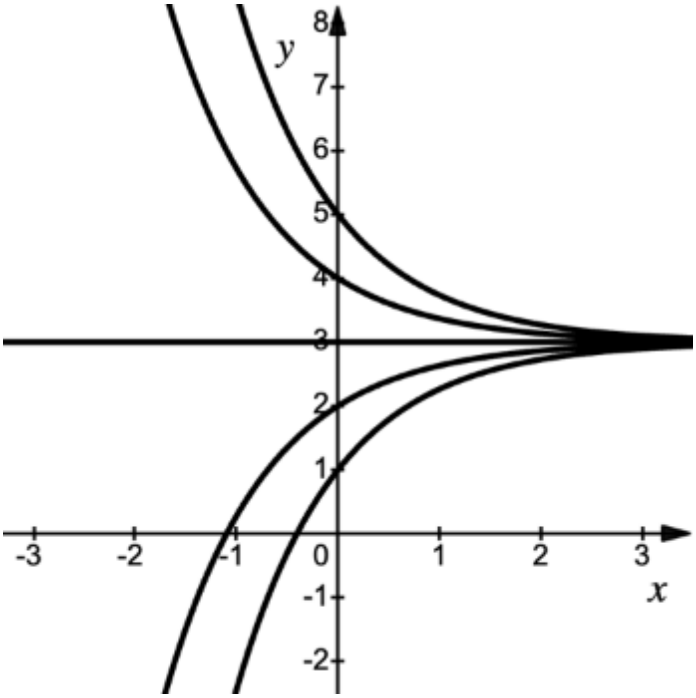
$$\frac{dy}{dx} + y = 3$$

Integrating Factor

$$= e^{\int 1\,dx} = e^x$$

$$\frac{dy}{dx} + y = 3$$
$$\Rightarrow e^x \frac{dy}{dx} + e^x y = 3e^x$$
$$\Rightarrow \frac{d}{dx}(e^x y) = 3e^x$$
$$\Rightarrow e^x y = 3e^x + c$$
$$\Rightarrow y = 3 + ce^{-x}$$

9 c)



10 a)

Trial:

$$y = k \Rightarrow \frac{dy}{dx} = 0$$
$$\therefore 0 - k = 2 \Rightarrow k = -2$$

So $y = -2$ is a constant solution of the differential equation.

10 b)

$$\frac{dy}{dx} - y = 2$$

Integrating Factor

$$= e^{\int -1\,dx} = e^{-x}$$

$$\frac{dy}{dx} - y = 2$$
$$\Rightarrow e^{-x}\frac{dy}{dx} - e^{-x}y = 2e^{-x}$$
$$\Rightarrow \frac{d}{dx}(e^{-x}y) = 2e^{-x}$$
$$\Rightarrow e^{-x}y = -2e^{-x} + c$$
$$\Rightarrow y = -2 + ce^{x}$$

10 c)

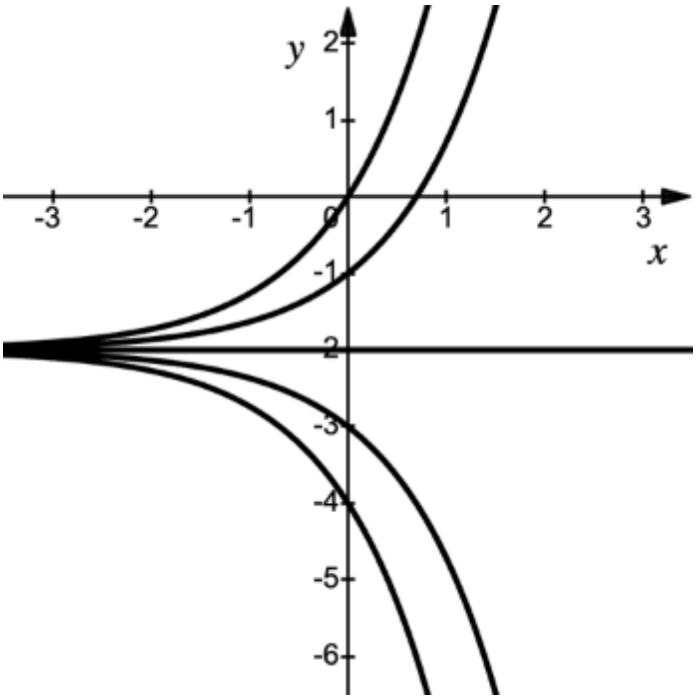
As $x \to \infty, e^x \to \infty$

So if $c > 0$, then $y \to \infty$

If $c = 0$, then $y = -2$

If $c < 0$, then $y \to -\infty$

10 d)



11 a)

Trial:

$$y = k \Rightarrow \frac{dy}{dx} = 0$$
$$\therefore 0 + 2k = 6 \Rightarrow k = 3$$

So $y = 3$ is a constant solution of the differential equation.

11 b)

$$\frac{dy}{dx} + 2y = 6$$

Integrating Factor

$$= e^{\int 2\,dx} = e^{2x}$$

$$\frac{dy}{dx} + 2y = 6$$
$$\Rightarrow e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$$
$$\Rightarrow \frac{d}{dx}(e^{2x}y) = 6e^{2x}$$
$$\Rightarrow e^{2x}y = 3e^{2x} + c$$
$$\Rightarrow y = 3 + ce^{-2x}$$

11 c)

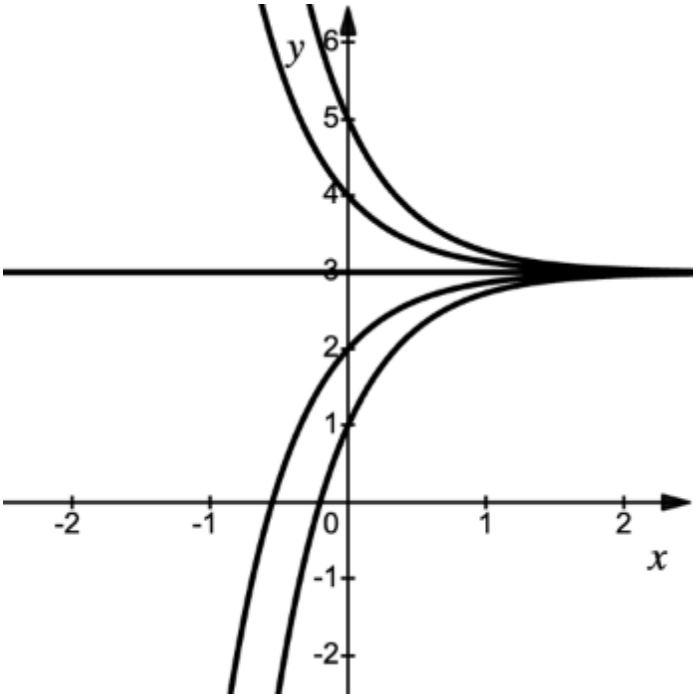
As $x \to \infty, e^{-2x} \to 0$

So if $c > 0$, then $y \to 3$

If $c = 0$, then $y = 3$

If $c < 0$, then $y \to 3$

11 d)



12 a)

Trial:

$$y = k \Rightarrow \frac{dy}{dx} = 0$$

$$\therefore 0 + 2x \times k = 4x \Rightarrow k = 2$$

So $y = 2$ is a constant solution of the differential equation.

$$\frac{dy}{dx} + 2xy = 4x$$

Integrating Factor

$$= e^{\int 2x\,dx} = e^{x^2}$$

$$\frac{dy}{dx} + 2xy = 4x$$

$$\Rightarrow e^{x^2}\frac{dy}{dx} + 2xe^{x^2}y = 4xe^{x^2}$$

$$\Rightarrow \frac{d}{dx}(e^{x^2}y) = 4xe^{x^2}$$

$$\Rightarrow e^{x^2}y = 2e^{x^2} + c$$

$$\Rightarrow y = 2 + ce^{-x^2}$$

12 b)

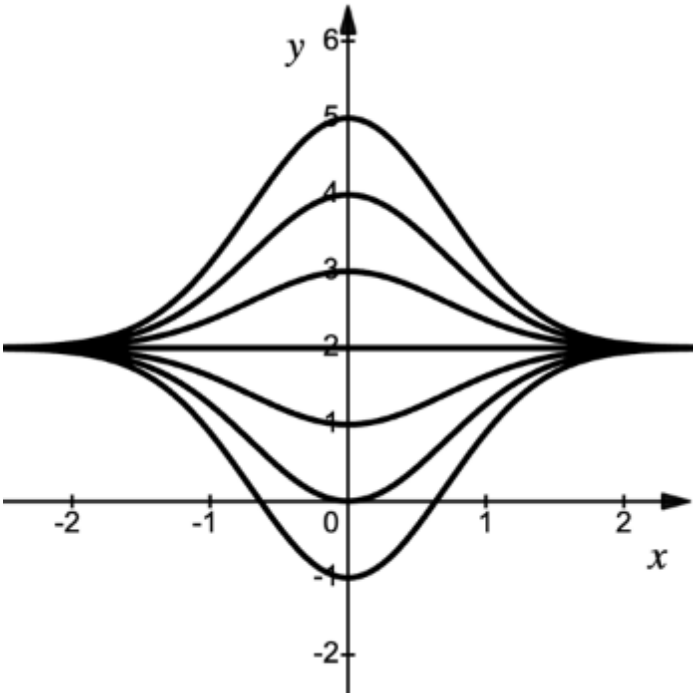
As $x \to \infty, e^{-x^2} \to 0$

So every solution curve tends to $y = 2$ as x tends to infinity

As $x \to -\infty, e^{-x^2} \to 0$

So every solution curve tends to $y = 2$ as x tends to negative infinity

12 c)



13 a)

$$\frac{dy}{dx} + y = x + 1$$

Integrating Factor

$$= e^{\int 1\,dx} = e^x$$

$$\frac{dy}{dx} + y = x + 1$$
$$\Rightarrow e^x \frac{dy}{dx} + e^x y = (x+1)e^x$$
$$\Rightarrow \frac{d}{dx}(e^x y) = (x+1)e^x$$

Use Integration by Parts:

$$u = x + 1, \quad \frac{dv}{dx} = e^x, \quad \frac{du}{dx} = 1, \quad v = e^x$$

$$\therefore \int (x+1)e^x \ln x\,dx = (x+1)e^x - \int e^x\,dx$$
$$= xe^x + e^x - e^x + c = xe^x + c$$

$$\therefore e^x y = xe^x + c$$
$$\Rightarrow y = x + ce^{-x}$$

13 b)

$y = x \Rightarrow \frac{dy}{dx} = 1$

So as $1 + x = x + 1, y = x$ is a solution of the differential equation.

13 c)

As $x \to \infty, e^{-x} \to 0$

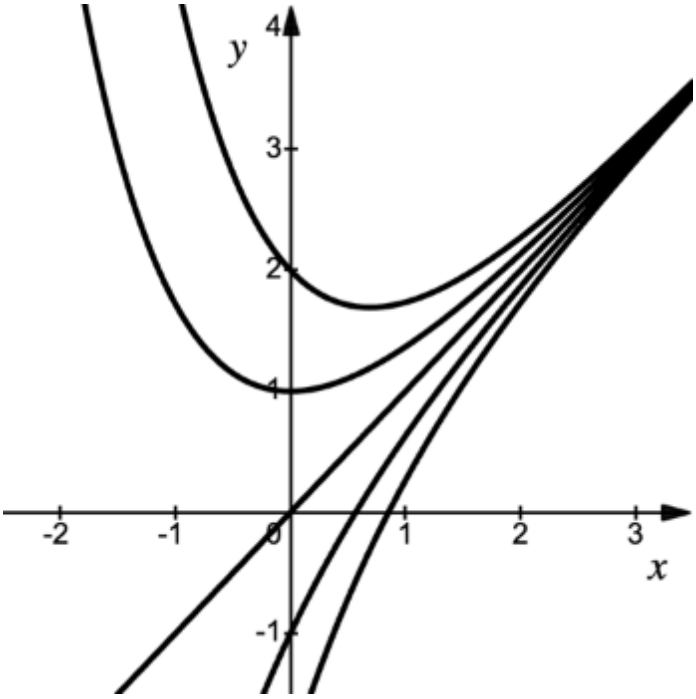
So if $c > 0$, then $y \to x$

If $c = 0$, then $y = x$

If $c < 0$, then $y \to x$

So every solution curve approaches the line $y = x$

13 d)



13 e)

Let one solution curve have equation $y = x + c_1 e^{-x}$ and let another solution curve have equation $y = x + c_2 e^{-x}$, where $c_1 \neq c_2$

If they were to intersect, then:

$$x + c_1 e^{-x} = x + c_2 e^{-x}$$
$$\Rightarrow c_1 e^{-x} = c_2 e^{-x}$$
$$\Rightarrow c_1 = c_2$$

But $c_1 \neq c_2$ so the solution curves cannot intersect.

14 a)

$$\frac{dy}{dx} - y = x$$

Integrating Factor

$$= e^{\int -1\,dx} = e^{-x}$$

$$\frac{dy}{dx} - y = x$$
$$\Rightarrow e^{-x}\frac{dy}{dx} - e^{-x}y = xe^{-x}$$
$$\Rightarrow \frac{d}{dx}(e^{-x}y) = xe^{-x}$$

Use Integration by Parts:

$$u = x, \quad \frac{dv}{dx} = e^{-x}, \quad \frac{du}{dx} = 1, \quad v = -e^{-x}$$

$$\therefore \int xe^{-x}\,dx = -xe^{-x} + \int e^{-x}\,dx$$
$$= -xe^{-x} - e^{-x} + c$$

$$\therefore e^{-x}y = -xe^{-x} - e^{-x} + c$$
$$\Rightarrow y = -x - 1 + ce^{x}$$

14 b)

$y = -x - 1 \Rightarrow \frac{dy}{dx} = -1$

So as $-1 - (-x - 1) = x, y = -x - 1$ is a solution of the differential equation.

14 c)

As $x \to -\infty, e^x \to 0$

So if $c > 0$, then $y \to -x - 1$

If $c = 0$, then $y = -x - 1$

If $c < 0$, then $y \to -x - 1$

So every solution curve approaches the line $y = -x - 1$

14 d)

As $x \to \infty, e^x \to \infty$

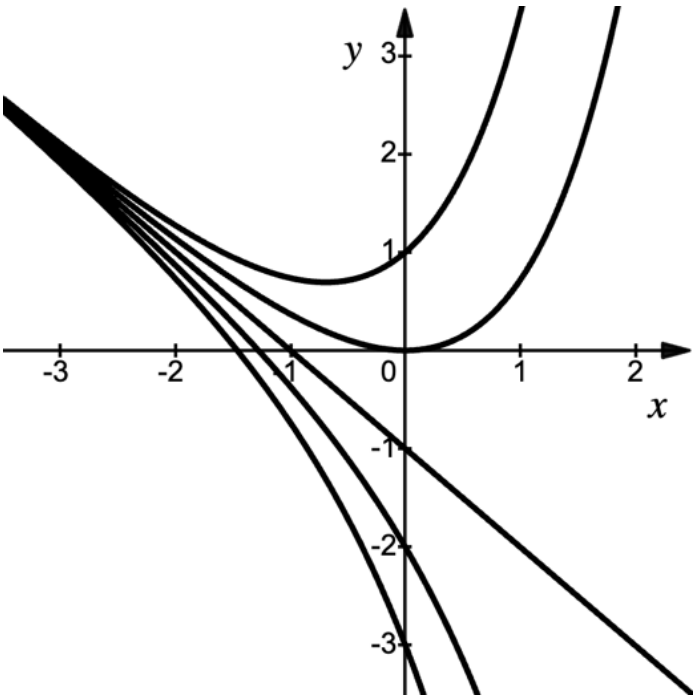
So if $c > 0$, then $y \to \infty$

If $c = 0$, then $y = -x - 1$

If $c < 0$, then $y \to -\infty$

So every solution curve diverges from the line $y = -x - 1$, unless $c = 0$

14 e)



15 a)

$$\frac{dy}{dx} + y = \sin x + \cos x$$

Integrating Factor

$$= e^{\int 1\,dx} = e^x$$

$$\begin{aligned} \frac{dy}{dx} + y &= \sin x + \cos x \\ \Rightarrow e^x \frac{dy}{dx} + e^x y &= e^x(\sin x + \cos x) \\ \Rightarrow \frac{d}{dx}(e^x y) &= e^x(\sin x + \cos x) \end{aligned}$$

Let $I = \int e^x(\sin x + \cos x)\,dx$

Use Integration by Parts:

$$u = \sin x + \cos x, \quad \frac{dv}{dx} = e^x, \quad \frac{du}{dx} = \cos x - \sin x, \quad v = e^x$$

$$\therefore I = e^x(\sin x + \cos x) - \int e^x(\cos x - \sin x)\,dx$$

Use Integration by Parts again:

$$u = \cos x - \sin x, \quad \frac{dv}{dx} = e^x, \quad \frac{du}{dx} = -\sin x - \cos x, \quad v = e^x$$

$$\begin{aligned} \therefore I &= e^x(\sin x + \cos x) - \left(e^x(\cos x - \sin x) - \int e^x(-\sin x - \cos x)\,dx\right) \\ \Rightarrow I &= e^x(\sin x + \cos x) - \left(e^x(\cos x - \sin x) + \int e^x(\sin x + \cos x)\,dx\right) \end{aligned}$$

$$\Rightarrow I = e^x(\sin x + \cos x) - e^x(\cos x - \sin x) - \int e^x(\sin x + \cos x)\,dx$$
$$\Rightarrow I = e^x \sin x + e^x \cos x - e^x \cos x + e^x \sin x - I$$
$$\Rightarrow 2I = 2e^x \sin x + c$$
$$\Rightarrow I = e^x \sin x + A$$
$$\therefore e^x y = e^x \sin x + A$$
$$\Rightarrow y = \sin x + Ae^{-x}$$

15 b)

As $x \to \infty, e^{-x} \to 0$

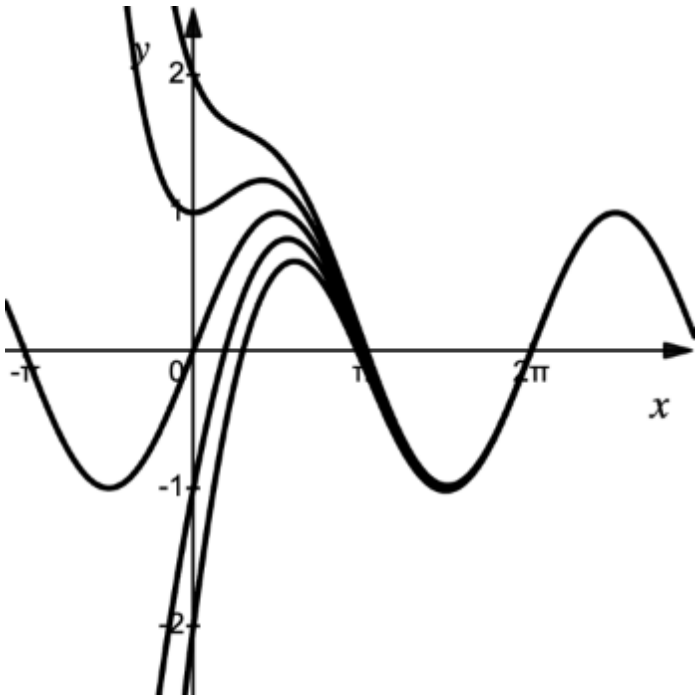
So if $A > 0$, then $y \to \sin x$

If $A = 0$, then $y = \sin x$

If $A < 0$, then $y \to \sin x$

So every solution curve approaches the curve $y = \sin x$

15 c)



16 a)

$$\frac{dy}{dx} + y = 4\sinh x$$

Integrating Factor

$$= e^{\int 1\,dx} = e^x$$

$$\frac{dy}{dx} + y = 4\sinh x$$
$$\Rightarrow e^x \frac{dy}{dx} + e^x y = 4e^x \sinh x$$

$$\Rightarrow \frac{d}{dx}(e^x y) = 4e^x \sinh x$$
$$\Rightarrow \frac{d}{dx}(e^x y) = 4e^x \times \frac{e^x - e^{-x}}{2}$$
$$\Rightarrow \frac{d}{dx}(e^x y) = 2e^x(e^x - e^{-x})$$
$$\Rightarrow \frac{d}{dx}(e^x y) = 2e^{2x} - 2$$
$$\Rightarrow e^x y = e^{2x} - 2x + A$$
$$\Rightarrow y = e^x + (A - 2x)e^{-x}$$

16 b)

$$y|_{x=0} = e^0 + (A - 2 \times 0)e^0$$
$$= 1 + A$$

So the solution curves intersect the y-axis at $(0, 1 + A)$

16 c)

As $x \to \infty, e^{-x} \to 0$ and $e^x \to \infty$

So every solution curve tends to infinity as x tends to infinity

As $x \to -\infty, e^{-x} \to \infty$ and $e^x \to 0$

So every solution curve tends to infinity as x tends to negative infinity

16 d)

$$y = e^x + (A - 2x)e^{-x}$$
$$\Rightarrow \frac{dy}{dx} = e^x - (A - 2x)e^{-x} - 2e^{-x}$$

Stationary points exist when $\frac{dy}{dx} = 0$

$$\therefore e^x - (A - 2x)e^{-x} - 2e^{-x} = 0$$
$$\Rightarrow e^x - Ae^{-x} + 2xe^{-x} - 2e^{-x} = 0$$
$$\Rightarrow e^{2x} - A + 2x - 2 = 0$$
$$\Rightarrow A = e^{2x} + 2x - 2$$

Substitute $A = e^{2x} + 2x - 2$ into $y = e^x + (A - 2x)e^{-x}$:

$$y = e^x + ((e^{2x} + 2x - 2) - 2x)e^{-x}$$
$$\Rightarrow y = e^x + (e^{2x} - 2)e^{-x}$$
$$\Rightarrow y = e^x + e^x - 2e^{-x}$$
$$\Rightarrow y = 2e^x - 2e^{-x}$$
$$\Rightarrow y = 4 \times \frac{e^x - e^{-x}}{2}$$
$$\Rightarrow y = 4 \sinh x$$

Alternatively:

The original differential equation is $\frac{dy}{dx} + y = 4\sinh x$

Stationary points exist when $\frac{dy}{dx} = 0$

$$\therefore 0 + y = 4\sinh x \Rightarrow y = 4\sinh x$$

16 e)

